BLESSING MAKAU

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PROFESSIONAL SUMMARY

Full-stack web developer who has completed the Software Engineering programme at Moringa School and is pursuing a Diploma in Computer Science at Zetech University (completing April 2027). Builds full-stack applications with Flask/SQLAlchemy backends and React frontends, covering REST API design, JWT authentication, relational data modeling, and cloud deployment. Comfortable working independently through the full project lifecycle from schema design and debugging to CI/CD and production deployment.

EDUCATION

Moringa School — Software Engineering Programme (Completed: 2026)

- Coursework: client-server architecture, SQL, SQLAlchemy, Marshmallow serialization, JWT authentication, deployment, CI/CD.

Zetech University — Diploma in Computer Science (Expected completion: April 2027 | Graduation: November 2027)

TECHNICAL SKILLS

Languages: Python, JavaScript, HTML, CSS, SQL

Backend: Flask, Flask-RESTful, SQLAlchemy, Flask-Migrate, Marshmallow, Flask-JWT-Extended, Gunicorn

Frontend: React, Vite

Databases: PostgreSQL, SQLite

Tools & Practices: Git/GitHub, GitHub Actions (CI/CD), Pipenv, Pytest, role-based access control, REST API design, M-Pesa payment integration, deployment (Render, Vercel)

PROJECTS

Tuinue Wasichana — *[tech stack, e.g. Flask, React, PostgreSQL, M-Pesa API]*

- Built a donation platform with an M-Pesa payment integration, enabling donations to be made through mobile money.
- Developed a dashboard that displays all donation information in one place [add specifics: totals, donor list, charts, filters].
- Deployed the application on [hosting services: frontend / backend / database].

Wardline (Hospital-System-V7) — *React + Vite, Flask, SQLAlchemy, Flask-JWT-Extended*

- Built a full-stack hospital management system covering eight clinical resources: patients, doctors, wards, admissions, discharges, appointments, prescriptions, and bills.
- Designed a generic CRUD blueprint factory to standardize REST endpoints across resources and implemented role-based access control via a custom decorator.
- Deployed the backend on Render (with Gunicorn) and the frontend on Vercel, resolving cross-origin routing and deployment configuration issues along the way.

Patient Record System — *Flask, SQLAlchemy, Flask-Migrate, Marshmallow, Flask-JWT-Extended, React*

- Developed a full-stack patient records application with JWT-authenticated API endpoints and serialized data models.
- Evolved the project from an earlier CLI-based version (Python OOP, JSON storage, modular package structure) into a full-stack web app.
- Managed the codebase through Git branching and merge workflows; hosted at github.com/BlexoC/patient-record-system.

CohortHub — *Flask-RESTful, SQLAlchemy, PostgreSQL*

- Built a bootcamp course-tracking platform demonstrating all three relational data patterns: one-to-one, one-to-many, and many-to-many (via an Enrollment join model).

- Wrote a full database seed script and used Flask-Migrate for schema migrations.

Flask Inventory Management System — *Flask, OpenFoodFacts API, Pytest*

- Built an inventory management tool integrating the OpenFoodFacts API for product data lookup.
- Wrote and verified 17 passing unit tests; refactored the codebase from four files into a leaner two-file structure (app.py, cli.py).